

The Excess Return Stayed at Home. The Defence Travelled Better

By Michael Trequattrini — Quantitative Researcher | Robust Systematic Trading & ML | Model Validation Specialist · LinkedIn post

The excess return stayed at home. The defence travelled better 🛡️

In the previous post, I discussed how inverse volatility and volatility targeting can create a more balanced portfolio core.

But improving the core does not eliminate the problem of market crises.

The 2008, covid and 2022 were not simply three versions of the same event.

- In 2008, treasuries protected the portfolio
- In 2022, they became one of its main sources of losses
- During covid, the main challenge was the speed of the shock.

For this reason, we did not rely on a single defensive signal.

I combined two different types of information:

- structural fragility, measured through equity-bond correlation, changes in that correlation and equity volatility
- the portfolio trend, measured by comparing the core portfolio's NAV with its moving average.

The final rule, called D3, does not try to decide which signal is correct.

It simply selects the more defensive exposure suggested by the two ✅

In the US development sample, the result looked very strong:

👉 maximum drawdown fell from approximately -16.2% to -9.6%.

But results from the market used to develop a strategy are only the beginning.

The real question was:

👉 Would the protection still work elsewhere, without changing the rules?

I therefore froze the specification and tested the mechanism across different markets.

In emerging markets, D3 reduced maximum drawdown from approximately -21% to -12.5%, while improving the Sortino ratio from 0.75 to 1.00.

By contrast, a more ambitious strategy designed to improve sharpe failed its test in developed markets outside the US.

This led to the most important conclusion of the research → defence transfers better than excess return 💡

This does not mean that a defensive rule will always work, or that it will work in every market.

It means that the ability to reduce deep losses appears to be more stable than the ability to consistently generate a higher CAGR or Sharpe ratio 📌

That distinction matters.

A strategy may not produce higher average returns, but it can still be more useful if it reduces the probability that investors abandon it at the worst possible moment.

The most robust result of the project was not a steeper equity curve. It was an equity curve that was harder to break 🇫🇷

Source: https://www.linkedin.com/posts/mich97_the-excess-return-stayed-at-home-the-defence-ugcPost-7485282498445340672-oE_s/